Milestone Three: Modeling Customer Conversion Intent

Digital Marketing Campaign Performance Analysis

Technical Report

Lauren Chagaris

Online Master's of Data Science, Boston University

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1. Executive Summary

IntentCX is a proposed customer routing system for digital marketing that sends customers who are likely to convert toward premium engagement resources and sends customers who are unlikely to convert toward a lower-cost path, with the goal of reducing wasted spend without losing sales. This project tested whether conversion can be predicted from customer behavior, using a dataset of 8,000 customers and a progression of regression, classification, and clustering methods. The central finding is that conversion is predictable, but only from what customers do rather than who they are. The best model, Gradient Boosting, reached an AUC of 0.8066, a measure of how well the model ranks likely converters above unlikely ones, and its strongest signals were prior purchases, pages viewed per visit, ad spend, and time on site. Demographic information such as age and income contributed almost nothing.

A second finding is that customers do not fall into natural behavioral groups that line up with conversion. Three different clustering methods produced groups that all converted at roughly the overall rate of 87.65 percent, which means a routing system cannot rely on simple customer segments and should instead score each customer with a trained prediction model. The recommended action is to pilot the Gradient Boosting model as a scoring layer, adjust its decision threshold to better identify low-intent customers, and validate the approach on an independent dataset before any production use. The likely impact is a lower engagement cost for each conversion, although the size of that saving has not yet been measured and depends on cost data the client would need to supply.

2. Introduction and Problem Definitions

2.1 Background and Domain Context

Digital marketing teams routinely spend on advertising, outreach, and engagement resources without a dependable way to tell which customers are likely to convert. The result is that premium resources are often directed at customers who would have converted anyway, while lower-intent customers absorb budget that produces no return. Industry practice addresses this through propensity modeling, in which historical behavior, demographics, and engagement data are used to score the likelihood that a customer will purchase or convert (CDP.com, 2026). Where customer interactions are handled by AI agents, the problem has an added cost dimension, because the compute required for a response varies with the capability of the model behind it, which turns the choice of which customer receives which tier of response into a direct cost decision.

IntentCX is the system proposed in this project to act on that idea. An intent classifier scores each customer, high-intent customers are routed to premium agents, and low-intent customers are routed to a cost-optimized path referred to as the Frugal Gateway. Because the Frugal Gateway reduces inference load for customers unlikely to convert, the marketing problem also connects to energy use in AI infrastructure, which is the longer-term motivation for the project.

The domain carries one structural difficulty that shapes every later decision. In this dataset, 87.65 percent of customers converted, so only about one customer in eight is a true low-intent case. A routing system gains value only if it can identify that minority reliably, and a model that simply predicts conversion for everyone would already be correct most of the time. Evaluation therefore has to look beyond accuracy.

2.2 Client Goals and Research Questions

The hypothetical client is a marketing operations team that wants to reduce cost per conversion by matching engagement resources to customer intent. Three research questions guide the technical work. The first asks whether conversion can be predicted from behavioral, demographic, and campaign features with enough discrimination to support tiered routing. The second asks which features carry the predictive signal and whether demographic features contribute at all. The third asks whether behavioral data forms natural customer segments that align with conversion when no outcome label is used to guide the analysis. These questions work toward lower cost per conversion because together they determine whether intent can be scored reliably enough to decide who receives costly resources, which inputs the score needs and whether sensitive attributes can be left out, and whether segmentation could supply routing tiers without an outcome label, so that premium spend is concentrated where it can change the result.

2.3 Success Criteria

Success is defined by three measurable conditions. A model should reach a held-out AUC-ROC of at least 0.75, a conventional threshold for acceptable discrimination. Its five-fold cross-validated AUC should fall within 0.03 of the held-out value, and the standard deviation across folds should not exceed 0.03, which shows that performance does not depend on one favorable split. The tolerance of 0.03 is smaller than the roughly 0.04 spread between the weakest and strongest classifiers in this project, so a gap of that size would be large enough to change how the models rank. Its errors should also be directionally useful for routing, meaning that few genuine converters are sent to the low-cost path. A full cost-based profit curve is outside the scope of this report because the dataset contains no cost data, and it is listed as a next step.

3. Data Overview

3.1 Source and Collection

The primary dataset is a digital marketing campaign dataset of 8,000 customer records. The data was obtained from a public Kaggle repository (zafarali27, n.d.). No collection method is documented, and the absence of missing values, the near-zero correlations between features and the outcome, and the uniform spread of the features suggest that the data is synthetic rather than observed from a live campaign. Each record describes one customer, with demographic attributes, campaign attributes, behavioral engagement measures, and a binary outcome indicating conversion.

3.2 Data Description and Dictionary

After removing the identifier column and encoding categorical fields, the modeling matrix contains 21 columns. The binary outcome, Conversion, equals 1 for 87.65 percent of customers. A continuous field, ConversionRate, was used as the target in the regression phase and is a campaign-level measure rather than the individual outcome. Age ranges from 18 to 69 (mean 43.6), Income from $20,014 to $149,986 (mean $84,664), AdSpend from $100 to $9,998 (mean $5,001), PagesPerVisit from 1.0 to 10.0 (mean 5.55), and TimeOnSite from 0.5 to 15.0 minutes (mean 7.73). The even spread of each feature across a bounded range, with means near the midpoint, is consistent with data generated from uniform distributions. Table 1 defines the columns most relevant to the analysis.

Table 1. Data dictionary for analysis-relevant columns

Column	Type	Meaning
Age	Numeric	Customer age in years

Gender	Categorical	Customer gender, encoded as a binary indicator
Income	Numeric	Customer income
CampaignChannel	Categorical	Acquisition channel (for example SEO, Referral, PPC, Email, Social)
CampaignType	Categorical	Campaign objective (for example Conversion, Consideration, Retention)
AdSpend	Numeric	Advertising spend attributed to the customer
ClickThroughRate	Numeric	Share of ad impressions that produced a click
WebsiteVisits	Numeric	Number of website visits
PagesPerVisit	Numeric	Average pages viewed per visit
TimeOnSite	Numeric	Average time spent on the site per visit
SocialShares	Numeric	Number of social shares
EmailOpens, EmailClicks	Numeric	Email opens and email link clicks
PreviousPurchases, LoyaltyPoints	Numeric	Prior purchase count and accumulated loyalty points
ConversionRate	Numeric	Campaign-level conversion rate, target in the regression phase
Conversion	Binary	1 if the customer converted, target in the classification phase

3.3 Data Quality Assessment

The dataset contained no missing values and no duplicate records. The dominant quality issue is class imbalance rather than missing data, since the 87.65 percent conversion rate limits how much any model can separate customers and makes accuracy a misleading metric. The dataset also comes from a single source, which means that findings have not been tested on a second population. Earlier project work in Module B found that behavioral patterns learned on consumer data did not transfer to a business-to-business dataset, with AUC falling to 0.6136, so generalization beyond this dataset should not be assumed.

3.4 Ethical Considerations

Three ethical considerations apply. First, the dataset includes demographic attributes, and a routing system that treats customers differently could disadvantage groups if those attributes influence the score. The modeling results are reassuring on this point, since demographic features carried negligible importance, which supports excluding them from any deployed model. Second, tiered routing means that some customers receive a cheaper level of service, so the system should be audited to confirm that low-intent assignments do not fall disproportionately on particular groups. Third, behavioral measures such as time on site and pages per visit imply tracking, so a deployment would need documented customer consent and data-handling practices consistent with applicable privacy rules.

4. Data Cleaning, Preprocessing, and Exploratory Data Analysis

4.1 Cleaning and Preprocessing

Because no values were missing and no records were duplicated, no imputation or row removal was required. Exploratory review found engagement features that were broadly spread without pronounced outliers or extreme skew, which meant that standard scaling was sufficient and no additional transformation such as log scaling or winsorization was required.

Categorical fields (Gender, CampaignChannel, and CampaignType) were one-hot encoded with the first level dropped to avoid perfect collinearity, and the encoding was cast to integer type to avoid boolean dtype errors in downstream libraries. The customer identifier was removed because it is a unique label for each row. It carries no behavioral information, and a flexible model could use it to memorize individual customers instead of learning patterns that generalize. Methods that depend on feature magnitude, including the regularized regressions,

logistic regression, support vector machines, K-nearest neighbors, and all clustering methods, used standardized features with mean zero and unit variance. Tree-based methods do not depend on scale and are unaffected by this step.

Two targets were defined. The continuous ConversionRate served the regression phase, and the binary Conversion served the classification phase and was held out entirely during clustering, then reintroduced afterward to profile the clusters. Keeping the label out of clustering allowed the unsupervised methods to be tested as independent evidence for the behavioral signal found by the supervised models.

4.2 Feature Engineering and Dimensionality Reduction

Feature engineering centered on interaction and polynomial terms in the first modeling week, added to test whether the weak linear fit was caused by missing non-linear structure. Multicollinearity was assessed with variance inflation factors. The highest values were 1.65 for CampaignChannel_Referral, 1.64 for CampaignChannel_PPC, and 1.61 for CampaignChannel_SEO, while every numeric feature stayed near 1.0. All values fall well below the common cutoffs of 5 and 10, so no feature was dropped. The modest inflation among the channel indicators is expected, because one-hot columns created from the same categorical field are mechanically related. Dimensionality reduction was applied through principal component regression and partial least squares regression. Neither approach lifted R-squared meaningfully above zero in the configurations tested. Together with the low variance inflation factors, this indicated that the weakness of the linear models was not caused by redundant predictors, so the regression target was set aside rather than engineered further.

4.3 Exploratory Visualizations and Key Findings

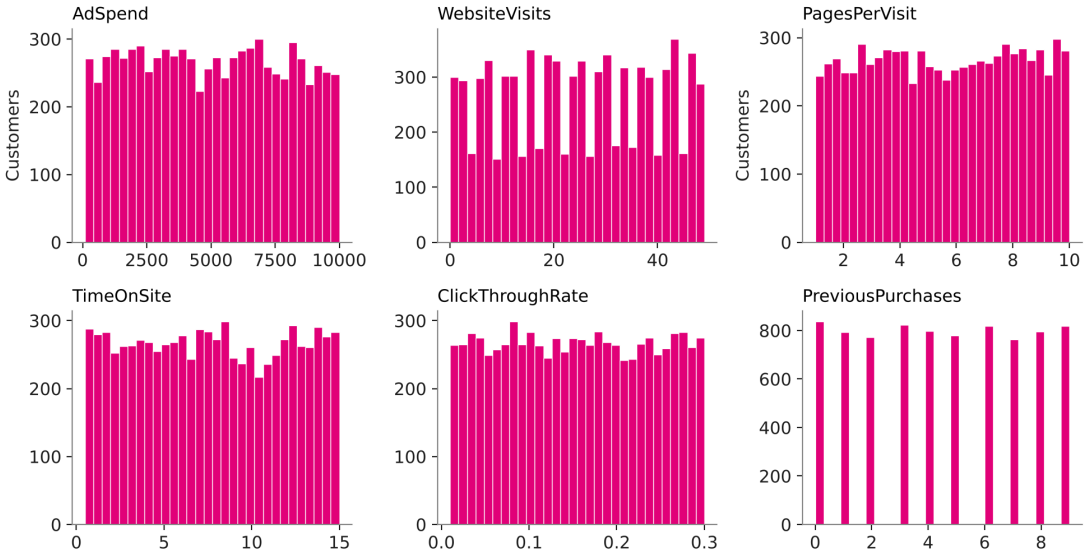


Figure 1. Distributions of key engagement features

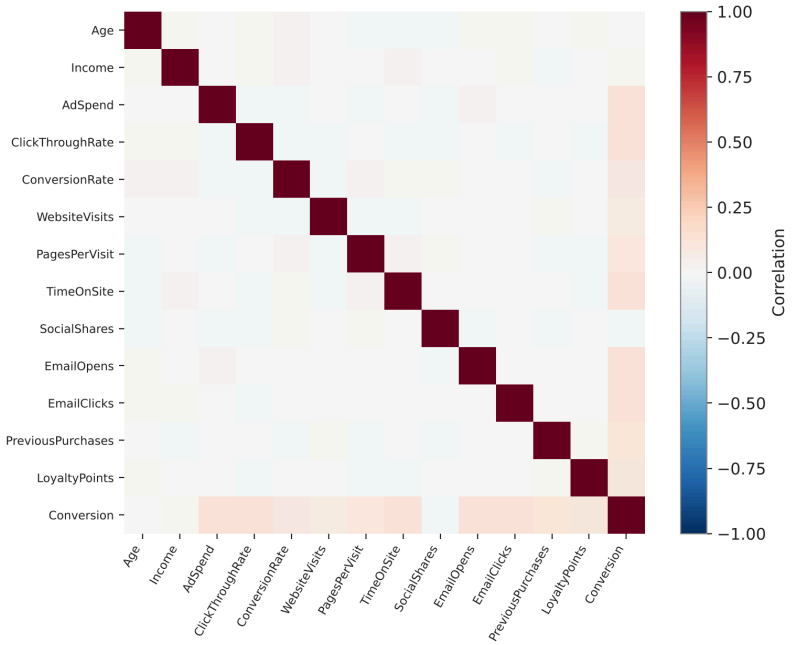


Figure 2. Correlation matrix of numeric features

Four findings from exploration shaped the modeling. First, the continuous ConversionRate was not linearly predictable from the available features, with R-squared near zero across every regression variant and with Lasso shrinking every coefficient to zero. Second,

the heavy class imbalance in the binary outcome made AUC-ROC the appropriate primary metric. Third, behavioral engagement features and demographic features behaved differently, with the engagement features showing the relationships that later models exploited. Fourth, the absence of strong linear structure suggested that tree-based methods capable of capturing feature interactions were the logical next step, and that suggestion was borne out by the classification results.

5. Modeling and Analysis

5.1 Model Choices

Models were chosen to span the main families of supervised and unsupervised methods, with each family contributing evidence about the same question. The regression phase used ordinary least squares with interaction and polynomial terms, Ridge, Lasso, ElasticNet, forward and backward feature selection, principal component regression, and partial least squares regression, all targeting ConversionRate. The classification phase used logistic regression, a linear support vector machine, Random Forest, K-nearest neighbors, and Gradient Boosting, all targeting Conversion. The unsupervised phase used K-Means, DBSCAN, and hierarchical agglomerative clustering. The progression was deliberate: linear models established a baseline, tree ensembles tested for interaction effects, and clustering tested whether the signal exists as natural structure.

5.2 Training Process and Data Splitting

Classification models were trained on 80 percent of the data and evaluated on a held-out 20 percent test set of 1,600 customers. The split was stratified by conversion so that the test set

preserved the 87.65 percent base rate, with 1,402 converters and 198 non-converters. Five-fold cross-validation was used to check that performance was stable across partitions rather than tied to one split. The unsupervised methods have no train and test split, so stability was assessed instead by refitting K-Means across five random seeds and by comparing multiple parameter settings for DBSCAN and multiple linkage methods for hierarchical clustering.

5.3 Evaluation Metrics

AUC-ROC was the primary classification metric because it measures ranking ability across all decision thresholds, and because accuracy would reward a model for predicting conversion for every customer in a dataset where 87.65 percent convert. Precision, recall, F1, and the confusion matrix supplemented AUC for the final model so that the direction of its errors could be examined. R-squared served the regression phase, and silhouette score served the clustering phase, measuring how well separated each cluster is from its nearest neighbor on a scale from negative one to positive one.

5.4 Hyperparameter Tuning

K-nearest neighbors was tuned across distance metrics and neighbor counts, settling on Manhattan distance, k equal to 101, and distance-weighted voting. A large k trades flexibility for stability, which suits a noisy outcome. Gradient Boosting received the most extensive tuning, through single-parameter sweeps of learning rate, tree depth, and number of estimators, followed by a joint grid search. The learning rate sweep showed the overfitting risk directly: as the rate rose from 0.01 to 0.5, training AUC climbed from 0.8454 to 0.9983 while test AUC stagnated near 0.79, widening the train-test gap from 0.0567 to 0.2095 (Figure 3). This matches documented behavior, in which learning rate controls overfitting through shrinkage (scikit-learn

Developers, n.d.). A depth sweep showed the same pattern, with depth 2 trees generalizing best at a gap of 0.0452 and depth 5 trees reaching a training AUC of 0.9876 against a test AUC of 0.7952, consistent with guidance that boosted trees should stay shallow (INRIA, n.d.).

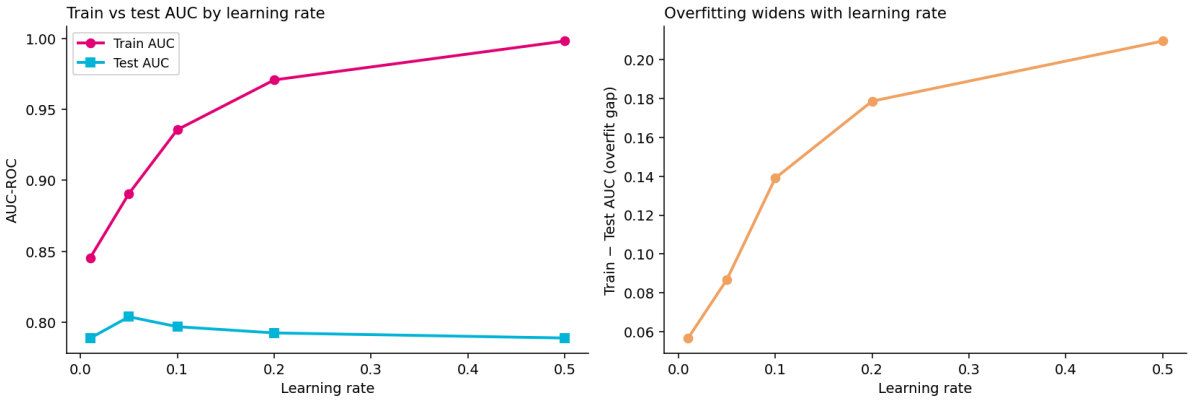


Figure 3. Gradient Boosting: Learning Rate Effect on Overfitting

The joint grid search selected a learning rate of 0.1, a maximum depth of 2, and 100 estimators. The selected learning rate exceeds the 0.05 that performed best in the isolated sweep, because the shallow trees limit each tree's capacity to overfit and allow a higher rate. Single-parameter sweeps built intuition, but only joint tuning located the optimum. DBSCAN's epsilon was selected from a k-distance graph at the point of maximum curvature (Mullin, 2020; DataCamp, 2026) and then tested across a range of values. K-Means used the elbow method and silhouette score together to select k equal to 4, and hierarchical clustering compared four linkage methods before selecting Ward linkage with three clusters.

5.5 Documentation of Modeling Decisions

Every model reported in this document was run and kept, including those that failed to produce signal. The regression results are retained because their failure is informative: a near-zero R-squared across nine linear approaches shows that the continuous campaign-level

target carries almost no learnable signal from these features, which is what motivated the move to the individual binary outcome. Full code for each modeling stage is available in the project notebooks at https://github.com/Data-Laur/capstone-assignments.

6. Results

6.1 Model Comparison

Gradient Boosting performed best, reaching a held-out AUC of 0.8066 and a five-fold cross-validated AUC of 0.8280 with a standard deviation of 0.0244. It exceeded Random Forest at 0.7976 and every earlier model, and it satisfied all three success criteria in Section 2.3. Its held-out AUC cleared the 0.75 threshold, its cross-validated AUC sat 0.0214 from the held-out value and so inside the 0.03 tolerance, and its fold standard deviation of 0.0244 fell inside the 0.03 limit. Recall of 0.9943 also met the routing requirement that few genuine converters be sent to the low-cost path. The improvement over the logistic regression baseline of 0.7708 is modest at about 0.036 AUC, which indicates that most of the available signal is captured by simple models and that non-linear methods refine it rather than transform it. Table 2 compares all classification and regression models.

Table 2. Supervised model comparison

Model	Target	Test AUC-ROC	Notes
Regression family (OLS, Ridge, Lasso, ElasticNet, PCR, PLSR)	ConversionRate	R-squared near 0	Lasso zeroed all coefficients
Logistic regression	Conversion	0.7708	Linear baseline
Linear SVM	Conversion	0.7654	Linear kernel, standardized features

K-nearest neighbors	Conversion	0.7680	Manhattan, k = 101, distance-weighted
Random Forest	Conversion	0.7976	Cross-validated AUC 0.8215
Gradient Boosting	Conversion	0.8066	Cross-validated AUC 0.8280, SD 0.0244

6.2 Feature Importance and Performance

The five most important features in the Gradient Boosting model were PreviousPurchases, PagesPerVisit, AdSpend, TimeOnSite, and ClickThroughRate, with importance values of roughly 0.14, 0.13, 0.12, 0.12, and 0.11 (Figure 4). Every one of them is behavioral or contextual, while Income and SocialShares contributed close to nothing. This agrees with Random Forest, which reached the same ranking, and it answers the second research question: the predictive signal lives in engagement behavior, and demographics add essentially no information. For routing, that is a useful result, because it means a deployed score can be built without sensitive attributes at little cost to accuracy.

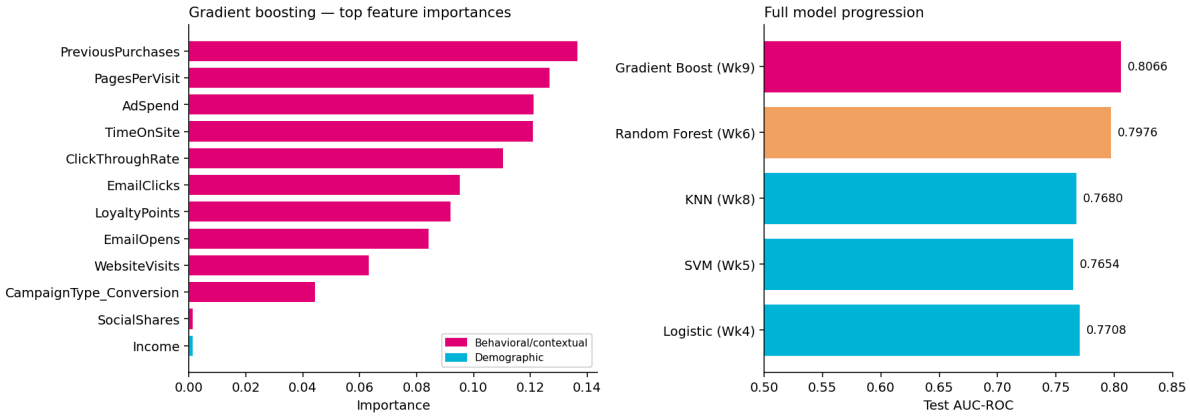


Figure 4. Gradient Boosting feature importance and model comparison

The confusion matrix for the final model showed 1,394 true positives, 49 true negatives, 149 false positives, and 8 false negatives, giving precision of 0.9034, recall of 0.9943, and an F1

score of 0.9467. Recall is very high, but specificity is low: only 49 of the 198 non-converters, about 24.7 percent, were identified correctly. The model therefore protects against sending genuine converters to the low-cost path, which is the more costly error, but it identifies only a quarter of the customers the Frugal Gateway is meant to capture.

6.3 Clustering Results

The unsupervised methods answered the third research question negatively. Table 3 summarizes them. K-Means selected k equal to 4 with a silhouette score of 0.1016, a weak value indicating overlapping clusters, and conversion rates across the four clusters ranged only from 86.8 to 88.3 percent. Centroid inspection showed that the clusters separated mainly by acquisition channel, the sharpest binary structure in the feature set, and not by engagement. Refitting across five seeds produced silhouette scores between 0.0998 and 0.1016, so the solution was stable but not meaningful for conversion.

Table 3. Clustering results

Method	Configuration	Silhouette	Conversion rate by cluster
K-Means	k = 4	0.1016	86.8% to 88.3%
DBSCAN	eps = 3.3, min samples = 10	-0.1118	89.4% (main cluster, n = 5,727) versus 83.2% (noise, n = 2,245)
Hierarchical (Ward)	k = 3	0.0810	87.3% to 88.2%

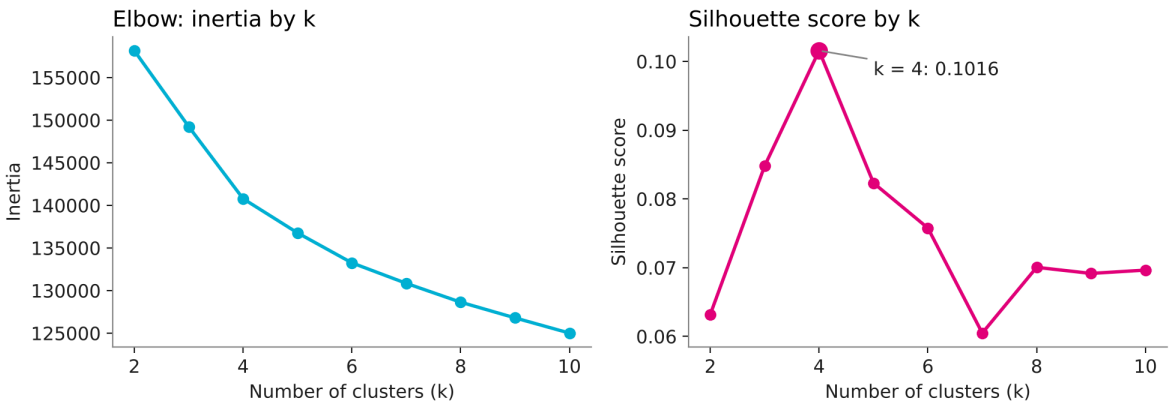


Figure 5. K-Means elbow and silhouette score by number of clusters



Figure 6. Conversion rate by K-Means cluster

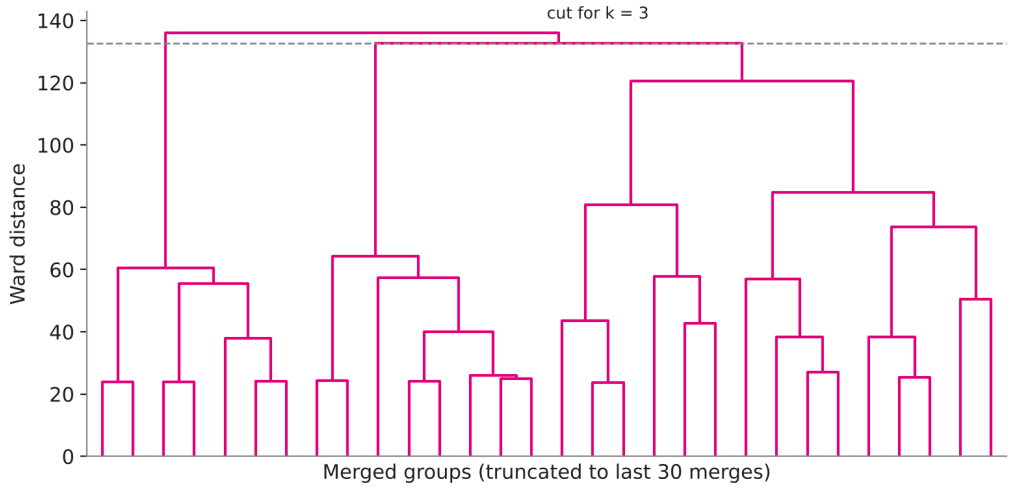


Figure 7. Ward linkage dendrogram (last 30 merges)

DBSCAN showed no stable middle ground. Values of epsilon below 3.2 fragmented the data into dozens of noise-dominated groups, while values above 3.35 collapsed it into one cluster. At the selected value, the five small clusters held fewer than ten points each and were excluded from interpretation, leaving a main cluster and a noise group of 28.1 percent of customers. The six-point gap in conversion between them is the largest separation any unsupervised method found, and it suggests that atypical customers convert somewhat less often, though the effect is small. Hierarchical clustering with Ward linkage again produced nearly equal conversion rates across clusters, and single linkage showed chaining behavior that made it unsuitable.

6.4 Limitations and Business Implications

Several limitations qualify these results. The high base rate caps what any partition can reveal, which is the most likely reason clustering failed to isolate a low-conversion group. Feature importance describes association and not causation, so a high importance for PreviousPurchases does not mean that raising purchases would raise conversion. The AUC of roughly 0.81 is moderate, and the low specificity means that the default threshold would under-route customers to the Frugal Gateway. All findings come from one dataset, and the earlier business-to-business test suggests transfer is not guaranteed.

For the client, the practical meaning is that a scoring model, not a segmentation scheme, is the right foundation for routing. Clusters did not predict conversion, whereas a trained classifier ranked customers well. Savings from routing will depend on how far the decision threshold can be moved to capture more low-intent customers without sacrificing recall, and that tradeoff can only be priced once the client provides cost figures.

7. Recommendations

The client should pilot the Gradient Boosting model as the scoring layer for tiered routing, using only behavioral and campaign features and leaving out demographic attributes, which add no predictive value and carry fairness risk. The pilot should begin as a shadow deployment in which the model scores customers but does not yet change routing, so predicted intent can be compared with actual conversion before any cost decision depends on it. The client should not build routing tiers from clustering, since none of the three methods produced groups that differ in conversion.

Implementation should address the weak specificity directly. Because the default threshold identifies only about a quarter of non-converters, the decision threshold should be tuned using the client's own costs for a premium interaction and for a missed conversion, which would also produce the profit curve that this report lacks. Model performance should be monitored for drift over time, and routing outcomes should be audited across customer groups to confirm that low-intent assignments are not concentrated unfairly. Latency and fallback behavior at the Frugal Gateway should be defined so that a misclassified high-intent customer can be escalated quickly.

Further analysis and data collection should focus on three areas. The approach should be validated on at least one independent dataset, given the failed transfer in earlier work. The client should collect cost and revenue data to enable a profit-based evaluation. More non-converter records, or targeted collection of low-intent behavior, would also help, since the scarcity of non-converters is the main constraint on the model's specificity. Calibration of predicted probabilities and cost-sensitive learning are the most promising modeling extensions.

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AI Appendix

Claude (Anthropic) was used throughout this capstone as a drafting and coding assistant. For this document, it was given the Milestone Three guidelines and rubric together with the student's own model outputs and results from earlier milestones, and it was asked to produce an outline organized by the required report sections, matching the format of the student's Milestone Two submission. The student supplied all results and numbers, reviewed the draft, revised it using feedback from a peer reviewer, and is responsible for verifying every figure against the notebooks. Claude also recomputed the variance inflation factors and regenerated Figures 1, 2, 5, 6, and 7 from the dataset file, and the clustering figures were checked against the silhouette scores and conversion rates from the Week 10 and Week 11 notebooks. Claude also provided notebook scaffolding and cell-by-cell interpretation guidance for the Week 10 and Week 11 clustering notebooks, and helped locate external sources for the references. Representative prompts included requests to map section lengths to rubric weights.